

Tic-Tac-Toe Game Using Minimax

AIM

To implement a **Tic-Tac-Toe game using the Minimax algorithm** in Python, where the computer selects the optimal move.

ALGORITHM

1. Start the game with an empty 3×3 board.
2. Player **X** and computer **O** take turns.
3. Check for a win, loss, or draw.
4. Generate all possible moves for the computer.
5. Apply Minimax to evaluate each move.
6. Select the move with the highest score for the computer.
7. Continue until a player wins or the game is a draw.
8. Display the result.
9. Stop.

CODE

```
# Tic-Tac-Toe using Minimax

board = [' ' for _ in range(9)]

def print_board():

    for i in range(0, 9, 3):

        print(board[i], '|', board[i+1], '|', board[i+2])

        if i < 6:

            print('--+---+--')

def winner(p):

    wins = [(0,1,2), (3,4,5), (6,7,8),

            (0,3,6), (1,4,7), (2,5,8),

            (0,4,8), (2,4,6)]

    return any(board[a] == board[b] == board[c] == p

               for a,b,c in wins)
```

```
def minimax(is_max):

    if winner('O'):

        return 1

    if winner('X'):

        return -1

    if ' ' not in board:

        return 0

scores = []

for i in range(9):

    if board[i] == ' ':

        board[i] = 'O' if is_max else 'X'

        scores.append(minimax(not is_max))

        board[i] = ' '

return max(scores) if is_max else min(scores)

def best_move():

    best_score = -2

    move = 0

    for i in range(9):

        if board[i] == ' ':

            board[i] = 'O'

            score = minimax(False)

            board[i] = ' '

    if score > best_score:

        best_score = score

        move = i
```

```
        return move

# Game

print("TIC-TAC-TOE")

print("You = X, Computer = O")

while True:

    print_board()

    pos = int(input("Enter position (1-9): ")) - 1

    if pos < 0 or pos > 8 or board[pos] != ' ':

        print("Invalid move!")

        continue

    board[pos] = 'X'

    if winner('X'):

        print_board()

        print("You Win!")

        break

    if ' ' not in board:

        print_board()

        print("Draw!")

        break

    move = best_move()

    board[move] = 'O'

    print("Computer chose:", move + 1)

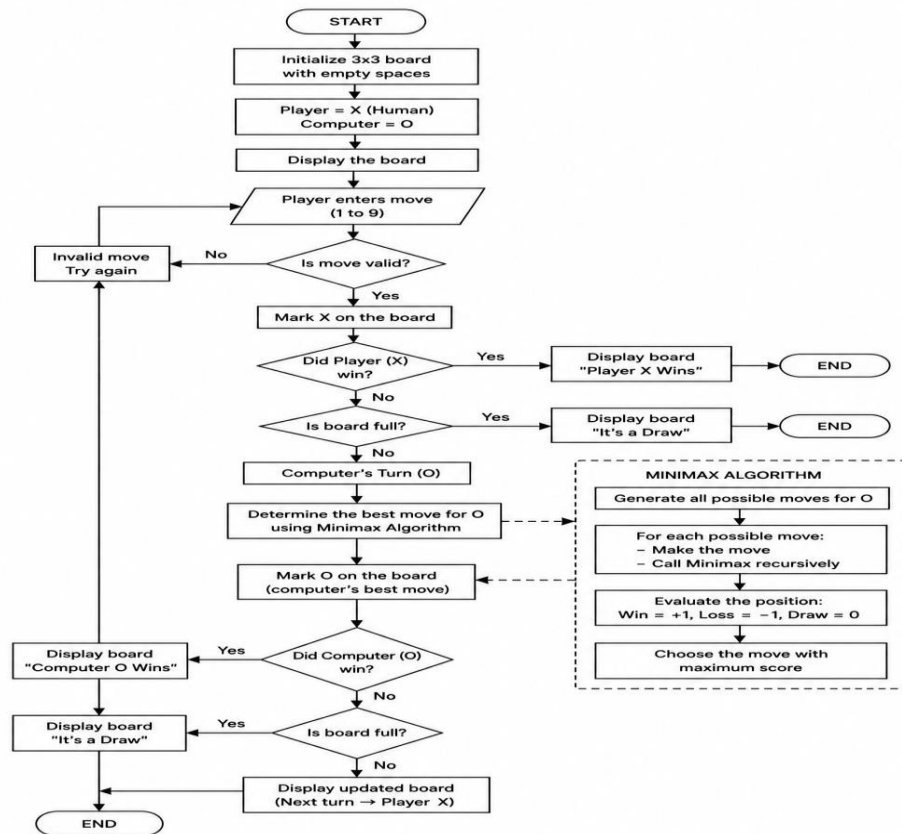
    if winner('O'):

        print_board()

        print("Computer Wins!")

        break
```

FLOW CHART



OUTPUT

```
PythonProject1 > TIC TAC TOE.py
X | | 
--+---+--
| 0 | 
--+---+--
| | 
Enter position (1-9): 3
Computer chose: 2
X | 0 | X
--+---+--
| 0 | 
--+---+--
| | 
Enter position (1-9): 7
Computer chose: 4
X | 0 | X
--+---+--
0 | 0 | 
--+---+--
X | | 
Enter position (1-9): |
```

The screenshot shows a terminal window with a dark background. It displays the output of a Python script named 'TIC TAC TOE.py'. The game board is represented by a 3x3 grid with characters X, O, and empty spaces. The terminal shows the player entering moves (3 and 7) and the computer responding with moves (2 and 4). The board state is updated after each move. The terminal window has a sidebar with various icons and a status bar at the bottom showing the current directory and file name.